Tewodros Seble

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Education

Addis Ababa University — Addis Ababa, Ethiopia
B.Sc. in Electrical and Computer Engineering;Microelectronics Specialization — *Sep 2019 – Jun 2024*

Statement

I am an Electrical and Computer Engineer with excellent professional and personal credentials. I am particularly interested in scientific and technology breakthroughs and their local implementation. I've worked as an intern, part-time employee, and volunteer in a variety of fields. As a result, I've improved my interpersonal and technological abilities. I've also improved transferable skills like critical thinking, problem solving, and flexibility. I am willing to advance my profession through networking with other persons and businesses. I am fascinated by artificial intelligence, robotics, and Circuit Design and Firmware development, and I am attempting to actually apply them to tackle social problems.

Skills

Personal: Communication, Project Management, Critical Thinking, Rational Decision Making, Analytic and Computational Skills, Complex Problem Solving,

Electrical and Electronics: Schematic and PCB Design, Embedded Systems and IoT, Digital and Analog Systems Design , Hardware description Language, Firmware Development, Webots, Remote Control systems, Signal Processing, Radio Waves and audio signals

Languages: C/C++, Java, Python, Scala, MATLAB, R

Technologies: Django, Node.js, DBMS Git Docker, AWS, Unity, Linux, Robot Operating System

Libraries: OpenCV, Scikit-Learn, PyTorch, Keras, TensorFlow, NumPy, Pandas, Matplotlib, Seaborn

Tools: Matlab and Simulink, Webots, KiCAD, EasyEDA, Altium, Quartus, Vivado, Eagle, Unity, IAR EW, LaTex

AI and Data Science: Data Preprocessing, Noise Filtering, Classification and regression system, Machine learning and deep learning model training and deployment, Recommender Systems, Object detection, Image segmentation, Image classification, Image generation, Neural network architecture and design(CNN, RNN), Automation and AutoML, Bias Control and Ethical AI

Experience

Purposeify — LA, USA, Remote
Machine Learning Engineer — *Nov 2023 – Present, Full time*

- The development and implementation of models capable of performing classification, recommendation, and regression tasks. involves preparing datasets, selecting and engineering features, choosing appropriate models, training the models on the data, and finally, deploying these models to production environments where they can provide insights or automate decisions.
- Adapt and refine large language models for specific applications. Large language models, like GPT (Generative Pretrained Transformer), are highly versatile but often require fine-tuning to tailor their responses or capabilities to specific industries, topics, or tasks. This fine-tuning process may involve training the models on specialized datasets to enhance their performance in particular domains or to ensure that their outputs align with organizational needs.

- Researching and applying different methodologies to optimize the accuracy and performance of AI algorithms. Exploring new architectures, training approaches, optimization algorithms, and other techniques that could potentially enhance the models' efficiency, reduce computational costs, or improve outcomes.

Addis-Tech Group Addis Ababa, Ethiopia(Internship)

Hardware Engineer *Sep 2023 – Feb 2024, Intern*

- Designed and tested schematic and PCB layouts for electronic circuits. This involved creating and validating the electrical circuit designs, then translating them into printed circuit board (PCB) layouts.
- Assembled and created devices by developing firmware using a variety of programming languages and tools, including C, C++, VHDL, Arduino, and Python. This task included not only the hardware design but also the development of the software that controlled and interacted with the hardware components. Firmware was written and tested to ensure proper operation of electronic devices
- Assembled and programmed microcontrollers and microprocessors. This entailed the physical assembly of these integrated circuits onto PCBs and configuring them through programming to perform specific functions

Ethiopian Health Professional Students Association Addis Ababa, Ethiopia

Project Team Leader *Jun 2023 – Nov 2023,*

- Participated as an E-learning project team leader for Health professionals learning platform.
- Assigned tasks for team members and give elaboration on topics, test and deploy deliverables from team members.

School of Marvel International Asella, Ethiopia

Programming trainer *Jul 2022 – Sep 2022,*

- Prepared modules for introduction to programming with python for highschool students.
- Given practical lessons on operations, functions, objects, classes and other basic programming contents.

Nile-Tech Engineering PLC Addis Ababa, Ethiopia (Part-time)

Embedded Systems Developer *Feb 2022 – May 2022*

- When I was at Nile-Tech Engineering PLC I have also worked on Embedded systems development.some of the projects are : Attendance Management system(using barcode and fingerprint) Temperature controlled fan system Smart home monitoring system(Sample project)

Asella STEM Center Asella, Ethiopia

Robotics Trainer *Jan 2021 – Mar, 2021*

- During the covid outbreak me and my former teachers got an idea of establishing a STEM center to train robotic and programming for students and university graduates. During that time I have successfully provided robotics training for about three months. our training has been published on Tikivah magazine.Link to News

School of Marvel International Asella , Ethiopia

Systems Developer *Jun 2020 – Oct 2020, Volunteer*

- Participated in the school's technological training and system digitization initiative as an application software developer.
- Developed the schools Student management system and Online learning platform. a web application built with django that can handle from student registration to examinations and result processing.
- Built a payroll management software that can calculate the monthly payroll of each teachers, managers and administrative workers by calculating the attendance, tax and other financial data.
- Developed attendance tracking system that uses finger print as an input and register it in the database using raspberry pi.

Participation

AfriconAI Best Project Award -Addis Ababa, Ethiopia

Winners *Oct 2024*

- Won First Place at the AfriconAI 2024 conference for our project, recognized as the Best Project Award winner.
- Exhibited the project to government officials, researchers, and distinguished guests at the event.

Bright Ethiopia Hackathon 2024 -Addis Ababa, Ethiopia

Winners *Mar 2024 –, Winner*

- I have participated on Bright Ethiopia Hackathon 2024 with 2 of my friends and we have secured 2nd place.
- For the hackathon we have worked on AI powered financial advisor and real time customer support using English, Afaan Oromoo and Amharic languages.

US Embassy Good Health and Well-Being -Remote

Developer *Nov 2020 –, Participation*

- Worked on an AI Powered android app and web app to predict liver disease based on MRI images of liver and give recommendation based on the classified diseases.
- Included features such as Drugs information and Liver disease prevention and treatment techniques.

AR/VR Africa Metathon -Addis Ababa, Ethiopia

Developer *Dec 2022 –, Participation*

- Developed a Virtual Reality based electronics laboratory where students can make simple electronics circuits and make electrical measurements based on the provided steps.(prototype level)

International Microelectronics Assembly And Packaging Society -

Student Member *Feb 2023 –, Membership*

- As a member of the International Microelectronics Assembly and Packaging Society (IMAPS), I have the privilege of engaging in various seminars, workshops, and remote educational offerings. Additionally, my membership provides me with unrestricted access to an extensive repository of publications, journals, and articles dedicated to the field of microelectronics. This invaluable resource allows me to stay current with the latest research and developments in the microelectronics industry.

International Association of Engineers -

Member *Apr 2023 –, Membership*

- I am an active member of the International Association of Engineers (IAENG) and have recently joined three of its prominent societies:The IAENG Society of Artificial Intelligence,The IAENG Society of Electrical Engineering, The IAENG Society of Scientific Computing. As a member of these societies, I have the opportunity to access the latest news, research journals, articles, and participate in remote conferences. These resources are invaluable for keeping me well-informed and up-to-date in my field of study, allowing me to stay at the forefront of developments in the relevant disciplines.

Black in AI Society -

Member *Apr 2023 –, Membership*

- As an active member of the Black in AI Society, I am dedicated to its mission of democratizing access to artificial intelligence (AI) through enhancing the presence, visibility, and well-being of Black professionals and leaders in the AI domain. My involvement spans participating in diverse projects, engaging in talk shows, competing in AI competitions, and participating in workshops.

Omdena -

Local Chapter Lead *May 2022 –, Chapter Lead*

- I am honored to serve as the Ethiopian Local Chapter Lead in the Omdena project. My active involvement in various Omdena projects, with a particular focus on Artificial Intelligence and Deep Learning, has provided me with valuable experiences and insights. Through my contributions to these projects, I have significantly enhanced my skills in this domain, furthering my professional growth and expertise.

Projects

Implementing ML and DL models on web, android and edge devices for coffee plant monitoring | *Article*

- Collecting, organizing and processing data for the models training.
- Training deep learning or machine learning models for coffee ripeness classification and coffee disease identification separately.
- Developing a recommender system that gives actions recommendation based on the ripeness level and the type of identified disease.
- Optimizing the performance of the model and deploying it on Web app, android app and Edge device using Raspberrypi.

Car License Plate Recogntion | *GitHub*

- Signal processing project on a license plate recognition of a car using different techniques without using machine learning/ deep learning altorithms
- Using the techniques of Gray Scaling, Gaussian Blur, Fast Fourier Transform, Band pass filtering for noise reduction, Inverse fast Fourier transform and Characters recogntion using OCR.

Tomato Disease identification web app | *GitHub*

- Training a deep learning model to identify the type of disease on a tomato leaf.
- Exporting the trained model for deploying it on a web app using django backend and finalize the frontend using bootstrap.

Hand Written Digits Identification using deep learning | *GitHub*

- Recognizing handwritten numbers from 0 to 9 using simple deep learning CNN model.
- Accurately recognize the number with an accuracy of more than 97.9 percent with a simple CNN model which 288 thousand parameters.

Heart Disease classification Machine Learning and deep learning models training | *GitHub*

- Training several machine learning models using Logistic regression, K-nearest neighbors, SVM, Naive Bayes, Decision trees and random forest algorithms.
- Creating a deep learning sequential model with simple architecture and compare the model's accuracy using confusion matrix.

Home Automation System With Java and Arduino | *GitHub*

- Home Automation system using arduino microcontroller with java Graphical user interface.
- Includes features such as Authentication, turning on and off lights, closing and opening doors, Light intensity sensor based automatic light controlling, Security alarming based on human approaching and so on...

Electronic Voting Machine Using ARM7 Microcontoller | *GitHub*

- Digital Voting system using ARM7-TDMI Microcontoller programmed on IAR-Embedded Workbench and simulated on Proteus.
- The system includes features like receiving user votes through buttons for each candidate, counting the votes for each candidate, comparing the vote counts in real-time, and promptly declaring the winner along with the vote difference.

Sequence Detector Moore Machine using VHDL | *GitHub*

- FPGA Based Sequence Detector Moore Machine in which the outputs of the system only depends on the current states of the machine.
- A VHDL project using quartus prime to detect a 0 nex to 1 input sequence using Moore Finite state machine. This finite state machine has three states based on the input values.As any other finite state machine this machine has three logic parts Current state logic,Next state logic,Output logic

Universal Shift Register Using VHDL | *GitHub*

- A universal shift register can load a parallel data word and perform shifting in either direction. There are four operations: load, shift right, shift left and pause. A control signal, ctrl, specifies the desired operation.

Schematic and PCB Design for Motor Controller | *Upwork*

- This project involves creating the schematic and PCB layout for a DC motor controller. The controller is designed to take input from a 220V/50Hz AC wall outlet and convert it to a 12V DC supply. It uses a potentiometer to regulate the motor's speed.

Research and Publications

Deep Learning Powered Equalization with Autoencoders for Improved 5G Communication

- As the second author of the paper, I contributed to the machine learning model component of the project focused on autoencoder-based equalization for 5G channels.

Leveraging AI and LLMs for Personalized Learning Pathways and Teaching Practices in Multilingual,

- At the "AI in Education" conference, I prepared and presented research on the educational challenges and opportunities faced by students, showcasing the impacts.

AI-Enhanced Equalization For Wireless Communication Systems

- Motivated by a previous study which is autoencoder based equalization to add extra AI models to enhance the equalization of wireless communication systems

Relevant Coursework

Major coursework: Electrical Circuits, Digital System Design, Numerical Methods,Statistics and Probability,Applied Electronics I-II, Signals and Systems, Electromagnetic Field Theory, Network Synthesis and Analysis , Microcomputers and Interfacing, Digital Logic Design, Digital Signal Processing, Instrumentation, Object Oriented Programming, Principles of Electronic Design, Computer Architecture and Organization, Integrated Circuit Technology

Minor coursework: EM Waves and Guided Structures, Introduction to Control Systems, Introduction to Communication ,Applied Engineering Mathematics I, II, III, Applied Modern Physics

Courses and Certificates

University of California Jan 2023
Introduction to Internet of Things and Embedded Systems(Coursera) View Certificate

IBM Nov 2013
IBM Machine Learning Specialist- Advanced View Certificate

Connors State University Feb 2024
Ethics of Artificial Intelligence (AI) View Certificate

Udemy Apr 2022
Artificial Intelligence in The Classroom View Certificate

Lund University Apr 2022
AI, Business & the Future of Work View Certificate

Udemy Feb 2023
Data Science and Machine Learning Basic to Advanced View Certificate

EIT Digital Oct 2023
Advanced Recommender Systems(Coursera) View Certificate

Deep Learning AI Sep 2023
Neural Network and Deep Learning(Coursera) View Certificate

Udemy Jan 2023
Practical Cisco Networking in Cisco Packet Tracer View Certificate

Udemy Aug 2023
VLSI Design View Certificate

References

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